

Group 1: Project Report

Investigating Relationships Between Crime, Deprivation and Wellbeing Across England (2024-2025)

Introduction

At the outset of the project, group one agreed to investigate a mission-driven topic, and we discussed fields such as health, crime, and census data.

After discussion, we agreed to use two datasets primarily – Police Recorded Crime Data (England, 2024-2025) and English Indices of Deprivation (IMD) 2025. We also planned to introduce a third dataset that contained wellbeing data; we thought this would be interesting to compare against the deprivation data.

The primary aim of the project was to investigate whether patterns of recorded crime are associated with levels of deprivation and wellbeing. To achieve this, our team combined crime data with measures of deprivation and wellbeing, conducted exploratory data analysis, and produced a range of visualisations to identify regional patterns and trends.

The primary objective of the project was:

To investigate relationships between recorded crime, deprivation and wellbeing across England.

This report begins by introducing the background and research questions before describing the methodology used to source, prepare and analyse the datasets. It then explains the implementation approach, presents the key findings and concludes with the project's limitations, recommendations and suggestions for future work.

Background

Crime is shaped by wider social and economic conditions such as deprivation, inequality and community wellbeing. Understanding how these factors interact is essential for building a fuller picture of why crime varies across places and populations. This project explores these relationships by examining crime patterns alongside measures of deprivation and wellbeing across England.

To investigate these relationships, the project focused on the following research questions:

1. Is there a relationship between deprivation and recorded crime across England?
2. How do crime patterns vary between regions?

3. Can wellbeing indicators provide further context when interpreting crime trends?
4. What insights can be gained from examining specific regions in greater detail?

The target audience for this analysis includes local authorities, policymakers, community organisations and researchers interested in understanding crime within its broader social context. By exploring how crime, deprivation and wellbeing intersect, the project aims to generate insights that support more informed decision-making, enable more targeted interventions and ultimately contribute to improved outcomes for communities across England.

Methodology and Design

Question Framing

The project was initially inspired by exploring whether crime was associated with wider social indicators such as deprivation and wellbeing. Following discussion and exploration of publicly available datasets, the research question evolved into investigating relationships between recorded crime, deprivation and wellbeing across England.

Data Sourcing

Three datasets were selected for analysis:

Police Recorded Crime Data (England, 2024–2025)

This dataset was chosen as the primary source because it provides detailed information on recorded offences across police force areas and regions in England.

English Indices of Deprivation (IMD) 2025

The IMD dataset was selected to provide information about relative deprivation across England. The dataset includes both deprivation ranks and deciles, allowing areas to be compared nationally.

UK Measures of National Wellbeing

The wellbeing dataset was included to provide broader social context and to explore whether areas with different wellbeing outcomes also exhibit different crime patterns.

The datasets were obtained from official UK Government departments and the Office for National Statistics (ONS), ensuring that the analysis was based on reliable, publicly available data. Two additional supporting datasets—a Local Authority District lookup and mid-year population estimates—were used during preprocessing and comparative analysis.

Data Preparation

Data preprocessing included:

- Checking for missing values and duplicate records.

- Filtering datasets to relevant geographic regions (e.g. removing data outside England).
- Standardising column names and geographic identifiers.
- Renaming columns to enable dataset integration.

During initial exploration, it became apparent that the datasets used different geographic classifications, including Police Force Areas, Lower Layer Super Output Areas (LSOAs) and Local Authority Districts (LADs). To enable meaningful comparison across datasets, these geographic units were standardised where possible using ITL1 and ITL2 classifications. Additional lookup tables published by the Office for National Statistics (ONS) were used to support this process.

Exploratory analysis included reviewing dataset dimensions, data types, summary statistics, missing values, duplicates and category distributions before investigating regional patterns and relationships between variables.

Analysis Methodology

The analysis combined descriptive statistics, aggregation techniques, correlation analysis and data visualisation to investigate relationships between crime, deprivation and wellbeing across England.

Exploratory Data Analysis (EDA) was first conducted to understand the structure and characteristics of each dataset. This included calculating summary statistics, examining data distributions and identifying regional patterns. GroupBy operations were used extensively to aggregate crime data by ITL1 region, ITL2 region and offence category, allowing comparisons between geographic areas and offence types.

Visualisation techniques were chosen according to the analytical objective. Bar charts and scatter plots were used to compare regional patterns and relationships, heatmaps were used to explore correlations, and choropleth maps were used to visualise spatial patterns in deprivation across London.

Implementation

Team Contributions

The project was completed collaboratively, with team members contributing to different stages of the data science lifecycle, including data preparation, exploratory analysis, visualisation, machine learning and report writing. Key areas of investigation included:

- Crime exploratory data analysis.
- Crime and deprivation comparisons.
- Crime and wellbeing comparisons.
- Regional deep dives.

This division of responsibilities allowed the team to investigate multiple perspectives while contributing to a shared research question.

Tools and Technologies

The project was developed using **Python** within **Jupyter Notebook**. Data cleaning and manipulation were carried out using **Pandas**, numerical computations using **NumPy**, and data visualisation using **Matplotlib** and **Seaborn**.

Geographic analysis and choropleth mapping were performed using **GeoPandas**, while **Scikit-learn** was used to implement an exploratory machine learning model. GitHub was used for version control and collaborative development.

Visualisation Accessibility

Accessibility was considered throughout the design of the project's visualisations. Colour-blind-friendly palettes, including Okabe–Ito and Viridis, were selected where appropriate to improve readability and ensure that charts and maps remained accessible to a wider audience, including users with deuteranopia and protanopia. Where sequential colour schemes were more appropriate, single-colour palettes with varying shades were used to maintain visual clarity while preserving accessibility. These design choices align with recognised data visualisation best practices and support the clear communication of analytical findings.

Challenges and Adaptations

Our group identified datasets that we thought looked interesting and might work well together, and then tried to formulate a hypothesis to test that brought them together. In retrospect, we should have discussed a hypothesis before looking for the data. This would have spared us from new challenges later on.

An example of the challenges later on included differences in geographic granularity between datasets. While deprivation data was available at neighbourhood level, crime data was often reported at police force level. For the London crime and deprivation analysis at ITL2 level, the crime data was not available at a sufficiently detailed geographic level. As a result, the analysis pivoted towards examining deprivation patterns at multiple geographic scales and assessing how aggregation influences interpretation.

Agile Working Practices

Jade led the team as Project Manager. Findings from exploratory analysis informed subsequent questions and visualisations. Regular Slack Huddles and collaboration via DMs in Slack allowed progress to be reviewed, peers to give feedback, challenges discussed, and tasks to be assigned/reassigned when necessary.

Results

Crime Analysis

Exploratory analysis of the crime dataset revealed substantial variation in recorded offences across England. London recorded the highest overall number of offences among all ITL1 regions, while the North East recorded the lowest.

At offence group level, Violence Against the Person, which includes offences such as assault, stalking and harassment, represented the largest category of recorded crime nationally. At offence subgroup level, Violence Without Injury was the most frequently recorded offence type.

While broad offence patterns were relatively consistent across England, London exhibited a distinctive crime profile. Three of London's five most common offence subgroups were theft-related (Other Theft Offences, Theft from the Person and Shoplifting), suggesting that acquisitive crime plays a particularly important role within the capital. These findings justified selecting London for more detailed investigation alongside deprivation data.

Deprivation Analysis

Exploratory analysis of the English Indices of Multiple Deprivation (IMD) dataset identified considerable variation in relative deprivation across England. Average IMD rankings and deciles demonstrated clear differences between regions, highlighting areas experiencing higher and lower levels of deprivation.

To investigate these patterns further, London was selected as a case study following the national crime analysis. Initial aggregation at ITL2 level suggested broad regional differences in deprivation, with Inner London East appearing the most deprived and Outer London South the least deprived. However, neighbourhood-level mapping using Lower Layer Super Output Areas (LSOAs) revealed substantial variation within each region. This demonstrated that regional averages can conceal important local differences and highlighted the value of analysing deprivation at multiple geographic scales.

Wellbeing Analysis

Exploratory analysis of the wellbeing dataset identified several indicators that appeared to be associated with recorded crime. Correlation analysis suggested moderate relationships between crime levels and measures including life expectancy, life satisfaction and perceptions of safety, although the strength of these relationships varied between indicators.

These findings indicate that crime should be considered alongside wider measures of social wellbeing when examining regional differences. However, because the wellbeing dataset was available at a broader geographic level than the crime and deprivation datasets, the results should be interpreted as indicative rather than evidence of a causal relationship.

Integrated Findings

While each dataset provided valuable insights independently, the greatest value emerged when they were considered together. The crime dataset identified regional variation in recorded offences, the IMD dataset provided socioeconomic context, and the wellbeing dataset offered broader measures of quality of life. Examining these datasets together enabled a more comprehensive understanding of regional differences than would have been possible using a single dataset.

Although direct comparisons were sometimes constrained by differences in geography, several consistent patterns emerged. Regions with higher recorded crime often also exhibited higher levels of deprivation, while some wellbeing indicators showed moderate relationships with crime levels.

Conclusion

This project investigated relationships between crime, deprivation and wellbeing across England using multiple publicly available datasets.

The analysis identified substantial variation in crime levels between regions and highlighted London's distinctive crime profile. Deprivation analysis revealed substantial geographic variation across England. The London case study demonstrated that regional averages can conceal significant neighbourhood-level variation, highlighting the importance of analysing data at an appropriate geographic scale.

Limitations

Several limitations should be considered:

- Geographic boundaries were not fully aligned across datasets.
- Crime data was only available at police force level rather than neighbourhood level, limiting direct comparison with deprivation data.
- Deprivation data – measures relative deprivation rather than absolute deprivation, and is at LSOA level.
- Wellbeing data – this was available at a broader regional level, limiting direct comparison with the crime and deprivation datasets.

Recommendations

- Policymakers should consider neighbourhood-level variation when designing interventions.
- Future government datasets would benefit from using more consistent geographic boundaries to improve cross-dataset analysis.
- Future research could incorporate additional socioeconomic variables, such as employment, education or housing, to provide a more comprehensive understanding of the factors associated with crime.

Machine Learning Extension

As an exploratory extension, Jade and Rebecca (Parker) developed a linear regression model to investigate whether more frequently published crime data could be used to estimate future IMD rankings between official IMD releases.

Although preliminary, the model demonstrates how predictive techniques could support future monitoring where updated deprivation data is not yet available.

Given additional time, future work could extend the machine learning component by implementing multiple linear regression using a wider range of input features and evaluating feature importance. This would provide a more comprehensive understanding of the variables most strongly associated with deprivation. The evaluation metrics (R^2 , RMSE and MAE) established in this exploratory model also provide a baseline against which more advanced models could be compared and assessed.

References

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